FATEMEH RANJBARAN



Location: Bologna, Italy

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SUMMARY

Computer Vision Engineer experienced in developing and deploying deep learning models for industrial vision and robotics applications. Expertise in object detection, segmentation, and pose estimation, with hands-on experience building end-to-end machine learning pipelines using Python, PyTorch, and OpenCV.

PROFESSIONAL EXPERIENCE

AI & Computer Vision Engineer, *I.E.M.A Company, Bologna, Italy* *Sep 2025 - Present*

- Designed and deployed an end-to-end AI vision pipeline for robotic palletizing on edge devices.
- Implemented object detection, segmentation, and pose estimation for box localization in industrial environments.
- Developed camera–robot calibration pipeline enabling pixel-to-mm spatial mapping.
- Integrated AI perception with robot control via PLC / OPC-UA for autonomous pick-and-place execution.

AI Research Intern, *Mas-Tech srl (Tennis Federation), Bologna, Italy* *Nov 2023 – May 2024*

- Implemented deep learning models for automated event detection and action segmentation.
- Applied CNN-RNN and Transformer models for temporal action recognition.
- Optimized multi-loss functions to enhance model accuracy and robustness.

Python Developer with ML Focus, *FARABI Hospital, Tehran, Iran* *Jan 2015 – Sep 2021*

- Developed and implemented an HIS as a team (6 people), optimizing workflows, automating processes, and integrating AI tools for decision support, achieving 91% accuracy.
- Integrated machine learning and AI-driven tools into decision support systems.

EDUCATION

M.Sc. in Artificial Intelligence, University of Bologna, Italy *Oct 2021 – Mar 2025*

The first-ranked university in Italy according to the Times Higher Education ranking.

B.Sc. in Network Computer Engineering, University of Applied Science and Technology, Iran *Oct 2011–Jul 2015*

Graduated as the top-ranked student in the class.

PROJECTS

- **AI Vision Pipeline for Robotic Palletizing using Detection and Segmentation Models** – *Project at I.E.M.A*
- **Object Detection with Image and Text Queries** – *Project at I.E.M.A*
- **Action Segmentation in Tennis Matches Using Transformers** – *Master's Thesis (Internship)*
- **Analysed Breast Cancer Histology Images** – *Computer Vision course*
- **Image Deblurring** – *Deep Learning course*
- **Question Answering with Transformers on CoQA** – *NLP course*

SKILLS

Technical Skills

- **Programming:** Python (Pandas, NumPy, Matplotlib), C++
- **Machine Learning:** Scikit-learn
- **Deep Learning:** TensorFlow, PyTorch, Keras, Transformers
- **Computer Vision:** OpenCV, YOLO, Object detection,Segmentation

Soft Skills

- Teamwork and Collaboration
- Creativity
- Critical thinking
- Multitasking and Efficiency

LANGUAGE SKILLS

- **English:** Fluent
- **Italian:** A2 in progress
- **Persian:** Native